

Experiment 5: Clustering based data analytics using R/Python (K-Means, SOM algorithms)

Aim

To perform clustering on a dataset using the K-Means algorithm, determine the optimal number of clusters using the Elbow method, and evaluate clustering performance using the Silhouette Score.

Code

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler

# Load dataset
df = pd.read_csv("/content/CC GENERAL.csv")

# Drop ID column
df.drop("CUST_ID", axis=1, inplace=True)

# Handle missing values
df.fillna(df.median(numeric_only=True), inplace=True)

# Scaling
scaler = StandardScaler()
scaled_data = scaler.fit_transform(df)

import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

inertia = []
K = range(1, 11)

for k in K:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(scaled_data)
    inertia.append(kmeans.inertia_)

plt.figure()
plt.plot(K, inertia)
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Inertia")
plt.title("Elbow Method")
plt.show()

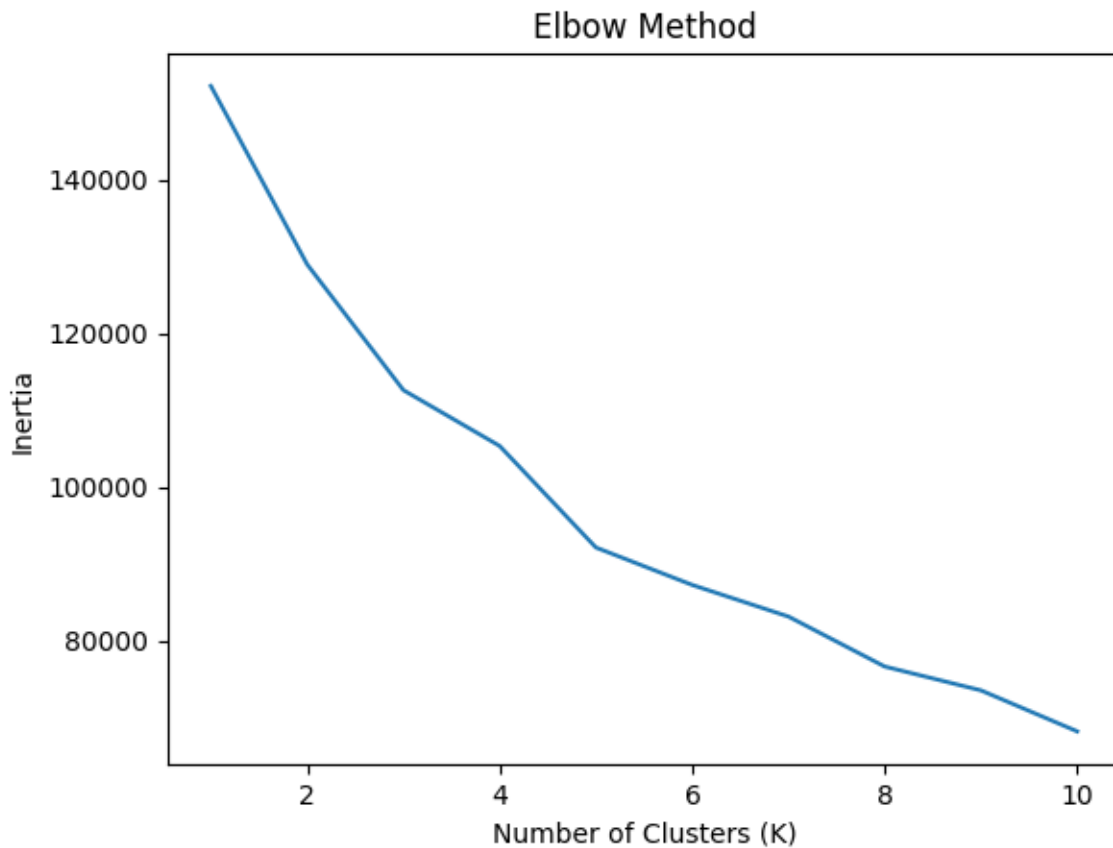
kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(scaled_data)
```

```
df["Cluster"] = clusters

from sklearn.metrics import silhouette_score

score = silhouette_score(scaled_data, clusters)
print("Silhouette Score:", score)
```

Output



Silhouette Score: 0.20671817817880617

Result

The dataset was preprocessed by removing unnecessary columns, handling missing values, and scaling the features. The Elbow method was used to determine the optimal number of clusters. K-Means clustering was applied to group the data into clusters, and the Silhouette Score was calculated to evaluate the quality of clustering. The results indicate that the data was successfully grouped into meaningful clusters.